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16 May 1988
SUPERSEDING
MIL-B-43154J
25 June 1986

MILITARY SPECIFICATION

BOOTS, HOT WEATHER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers a mildew resistant tropical combat boot that has a leather and nylon duck upper and a direct molded sole with a built-in spike resistant insole.

1.2 Classification. The boot shall be of one type in the sizes and widths as specified (see 6.2). The schedule of sizes is as listed in table I.

TABLE I. Schedule of sizes

Width	3	3½	4	4½	5	5½	6	6½	7	7½	8	8½	9	9½	10	10½	11	11½	12	12½	13	13½	14
XN	-	-	-	-	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
N	-	-	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
R	-	-	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W	-	-	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
XW	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8430

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

A-A-203	- Paper, Kraft, Wrapping
V-L-61	- Laces, Nylon
V-T-285	- Thread, Polyester
V-T-295	- Thread, Nylon
KK-I-570	- Insole, Footwear, Leather, Cattlehide
QQ-B-613	- Brass, Leaded and Nonleaded: Flat Products (Plate, Bar, Sheet, and Strip)
TT-C-490	- Cleaning Methods for Ferrous Surfaces and Pretreatments for Organic Coatings
DDD-L-20	- Label: For Clothing, Equipage, and Tentage, (General Use)
DDD-T-86	- Tape, Textile; Cotton, General Purpose (Unbleached, Bleached, or Dyed)
GGG-N-350	- Nippers and Pincers
PPP-B-636	- Boxes, Shipping, Fiberboard

MILITARY

MIL-I-1821	- Insoles, Footwear (Ventilating Type)
MIL-L-3122	- Leather, Cattlehide, for Footwear Uppers, Chrome Tanned, Fatliquored
MIL-T-5038	- Tape, Textile and Webbing, Textile, Reinforcing, Nylon
MIL-S-5059	- Steel, Corrosion-Resistant (18-8), Plate, Sheet and Strip
MIL-A-8625	- Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-C-13924	- Coating, Oxide, Black, for Ferrous Metals
MIL-W-17337	- Webbing, Textile, Woven Nylon
MIL-L-35078	- Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification For
MIL-C-41814	- Counter, Footwear
MIL-L-41815	- Lasts, Footwear, Combat, U.S., MIL-5
MIL-C-41836	- Cloth, Duck; Cotton and Nylon
MIL-C-43956	- Cloth, Twill, Cotton, 10.0 Ounces (339 g)

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STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods
FED-STD-311 - Leather, Methods of Sampling and Testing
FED-STD-601 - Rubber: Sampling and Testing
FED-STD-751 - Stitches, Seams, and Stitchings

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads
MIL-STD-731 - Quality of Wood Members for Containers and Pallets

(Copies of specifications, standards, and handbooks required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWING

U.S. ARMY NATICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

2-1-962 - Panama Tread Sole for Hot Weather; Boots; (Sole Dimensions)

(Copies of drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B 36 - Brass Plate, Sheet, Strip, and Rolled Bar
D 518 - Rubber Deterioration - Surface Cracking
D 573 - Rubber Deterioration in an Air Oven

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D 1052	- Rubber Deterioration - Cut Growth Using Ross Flexing Apparatus
D 1149	- Rubber Deterioration - Surface Ozone Cracking in a Chamber (Flat Specimens)
D 1630	- Rubber Property - Abrasion Resistance (NBS Abrader)
D 2240	- Rubber Property - Durometer Hardness
D 3951	- Standard Practice for Commercial Packaging
E 18	- Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials

(Copies should be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Guide samples. Samples, when furnished, are solely for guidance and information to the contractor (see 6.4). Variation from this document may appear in the sample, in which case this document shall take precedence.

3.2 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.3).

3.3 Materials. The materials shall conform to applicable documents and requirements specified herein. Recycled materials may be used, except for textile, leather, or rubber components (see 6.5).

3.3.1 Leather. All leather components shall contain not less than 0.18 percent nor more than 0.70 percent paranitrophenol fungicide.

* 3.3.1.1 Upper leather. The leather for the vamps, counterpockets, quarter front stays, and eyelet stays shall be smooth plated, full grain, or corrected grain side leather conforming to treatment A of MIL-L-3122, except that the thickness shall be as specified for the cut parts in table II. The break for the vamp shall be not more than a break pattern of 4 when examined as specified in 4.4.3. The color shall be black. All finishes applied by the boot manufacturer and tanner shall be compatible with one another to ensure a long-lasting low-gloss finish.

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TABLE II. Thickness of upper leather

Part	Thickness (ounces) Minimum	Maximum
Vamp	5	6
Outside counterpocket	4	6
Inside counterpocket	3 1/2	4
Quarter-front edge stays	3 1/2	4
Eyelet stays	3 1/2	4

3.3.1.2 Leather insoles. The leather insoles shall consist of two thicknesses: a 4-iron insole conforming to type I, tannage a or b, class 1 of KK-I-570 and a 3-iron insole conforming to type I or III, tannage a or b, class 1 of KK-I-570. Slaughter cuts not penetrating more than one-third of the thickness of the insole may be permitted in the 3-iron insole. Leather insoles split from a single unit shall conform to the above specified thicknesses after splitting. The insoles for this boot do not contain a channel area.

3.3.1.3 Counters. The counters shall conform to the requirements of MIL-C-41814 for Boots, Hot Weather.

3.3.2 Fabric and box toe materials.

3.3.2.1 Cotton and nylon duck. The cloth for the quarters and gussets shall be cotton and nylon duck conforming to MIL-C-41836. The color shall be Olive Green 107.

3.3.2.2 Backseam tape. The tape shall conform to type I, class 1, 1/2 or 17/32 inch wide of DDD-T-86.

3.3.2.3 Box toes. The box toes shall be cut from a material consisting of a woven base flannel or a processed (needled) fabric impregnated with a pyroxylin or polystyrene compound. The box toes may be either an activated or a prepared type. The thickness of this material shall be 0.0875 ± 0.0075 inch when tested as specified in 4.4.1. The box toes shall be of a suitable shape and size to engage the insole around its toe radius after lasting, and shall be skived 9/16 ± 1/8 inch wide at the breast.

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3.3.2.4 Vamp lining. The vamp lining shall be cut from cotton twill conforming to class 2 of MIL-C-43956.

3.3.2.5 Binding tape. The tape for the top edge of the gusset shall be nylon conforming to type III, class 1 or 2 of MIL-T-5038, except that the color shall be Olive Green 106 and the colorfastness shall be as specified in 3.3.2.5.1.

3.3.2.5.1 Colorfastness. The dyed tape shall show fastness to weathering equal to or better than the standard sample (see 6.4). When no standard sample is available, the dyed tape shall show "good" fastness to weathering when tested as specified in 4.4.1.

3.3.2.6 Quarter top facing, backstay, and ankle reinforcement tape. The quarter top facing and backstay tape shall be cut from 1-inch nylon webbing, and the ankle reinforcement tape shall be cut from 2-inch or two 1-inch pieces of nylon webbing conforming to class 1 or class 2 of MIL-W-17337, except that the webbing shall be a nylon 6,6, shall be dyed Olive Green 106, and shall show colorfastness to weathering equal to or better than the standard sample (see 6.4) when tested as specified in 4.4.1. When no standard sample is available, the dyed webbing shall show "good" fastness to weathering when tested as specified in 4.4.1.

3.3.3 Metal.

3.3.3.1 Steel shank. The steel shank shall be constructed from 19 gage, 0.0418 ± 0.0030-inch cold rolled carbon steel with a hardness ranging from 47 to 54 on the Rockwell C scale and shaped to conform to the arch of the MIL-5 last in MIL-L-41815. The width of the shank shall be 1 ± 1/32 inch. The shank shall be made with three ribs, and the overall height of each rib shall be 0.080 to 0.125 inch. Each rib shall taper off to end within 1 ± 1/8 inch from the toe end and within 1 ± 1/8 inch from the heel end. The shank shall have two prongs, each of 3/16-inch length, upward into the filler and two prongs, each of 3/32 to 5/32-inch length, downward into the insole. The steel shank shall have a zinc electroplated coating with dull or bright finish, or a zinc coating completely and uniformly applied to the base metal following a thorough cleansing by any method of TT-C-490. Steel shanks with black oxide coating conforming to class 1 of MIL-C-13924 or steel shanks with zinc phosphate coating may be used in lieu of shanks with zinc coating. The shank length grading shall be as specified in table III. Testing shall be as specified in 4.4.1.

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TABLE III. Shank lengths

Boots sizes (all widths)	3 to 6½	7 to 8½	9 to 10½	11 to 12½	13 to 14
Shank length in inches ± 1/32 inch tolerance	4-1/8	4-3/8	4-5/8	4-7/8	5-1/8

3.3.3.2 Tacks and staple wire. The heel seat lasting tacks shall be brass or steel. The brass tacks shall be cut from alloy No. 6 or 8 of ASTM B 36. The side lasting staples and toe lasting staples shall be steel. Tacks shall be of sufficient length to attach the parts through which they are driven and clinch onto the insole.

3.3.3.3 Eyelets. The eyelets shall be made of aluminum of 0.0160 ± 0.0015 inch thickness with roll setting barrel and shall conform to the following finished requirements when tested as specified in 4.4.1:

Outside diameter of flange - 0.480 to 0.496 inch
Outside diameter of barrel - 0.295 to 0.299 inch
Diameter of hole before setting - 0.235 to 0.240 inch
Overall length - 0.224 to 0.236 inch

* 3.3.3.3.1 Finish of eyelets. After fabrication, the eyelets shall be anodized in accordance with MIL-A-8625 and tumble coated with not less than two coats of black enamel.

3.3.3.4 Drainage eyelets and washers.

3.3.3.4.1 Drainage eyelets. The drainage eyelets shall be brass conforming to chemical composition copper alloy No. 260 of QQ-B-613. The drainage eyelet shall have a head with a diameter of 0.345 ± 0.005 inch, the outside diameter of the barrel shall be 0.218 ± 0.002 inch, and the barrel length shall be 0.185 ± 0.005 inch when tested as specified in 4.4.1.

3.3.3.4.1.1 Drainage eyelets (alternate). The drainage eyelets shall be brass conforming to chemical composition copper alloy No. 260 of QQ-B-613. The drainage eyelets shall have an outside diameter of 0.427 ± 0.005 inch, and the overall length shall be 0.086 ± 0.005 inch when tested as specified in 4.4.1.

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3.3.3.4.2 Washers. The washers shall be brass conforming to chemical composition copper alloy No. 260 of QQ-B-613. The washers shall have an outside diameter of 0.463 ± 0.005 inch, and inside diameter of 0.220 inch, +0.005, -0 inch, and a thickness of 0.009 ± 0.003 inch when tested as specified in 4.4.1.

3.3.3.4.2.1 Washers (alternate). The washers shall be brass conforming to chemical composition copper alloy No. 260 of QQ-B-613. The washers shall have a head with a diameter of 0.395 ± 0.005 inch, the outside diameter of the barrel shall be 0.287 ± 0.005 inch, the barrel length under the head shall be 0.185 ± 0.005 inch, and the thickness shall be 0.011 ± 0.003 inch when tested as specified in 4.4.1.

3.3.3.5 Steel insole plate. The steel for the insole plate shall be corrosion-resistant steel strip, composition 301, condition hard, thickness 0.011 ± 0.001 inch, 2D or 2B finish, conforming to MIL-S-5059.

3.3.4 Outersole and heel, rubber. The bottom assembly for both DMS biscuit and injection molding methods (rubber outersole and heel) shall have the following general characteristics and shall be tested as specified in 4.4.1 and 4.4.4:

a. Color: The color shall be black, nonmarking, producing only a slight erasable mark on white paper.

b. Volume swell: The volume swell shall be not greater than 60 percent in medium No. 6 fuel.

c. Abrasive index: The abrasive index shall be not less than 90 before and after accelerated aging for 70 hours at 100° ± 1°C.

d. Hardness: The durometer hardness shall be 70 ± 5. After accelerated aging for 70 hours at 100° ± 1°C, the durometer hardness shall change no more than 10 points from the original.

e. Cut growth: The cut growth, after accelerated aging for 70 hours at 100° ± 1°C, shall be no greater than 100 percent after 50,000 cycles.

f. Ozone resistance: There shall be no sign of cracking after the test for ozone resistance.

g. Bond strength: The bond strength shall be not less than 135 pounds except that one specimen out of the five tested may be below 135 pounds but shall be not less than 120 pounds. If any specimens in the sample show clear evidence of tearing into the leather or outersole rubber below 135 pounds, those specimens shall be discarded and new ones shall be selected for the test.

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3.3.4.1 Bottom design. The bottom design shall be bars and cleats conforming to Drawing 2-1-962. The sole thickness shall be as specified on the drawing and shall match within 1/16 inch between left and right boots.

3.3.5 Thread.

3.3.5.1 Thread, upper fitting. The thread for all upper fitting operations shall be bright nylon thread conforming to type I, II, or III of V-T-295, except that requirements for colorfastness shall not apply. The nylon thread shall be E for the needle and bobbin with the 301 stitch type, E for the needle, and B or E for the looper with the 401 stitch type. The color of the thread shall be black.

3.3.5.2 Thread, insole stitching. The thread for stitching the two layers of the leather insoles shall be a natural color polyester thread conforming to type I, class 1, subclass C, size 5, 3 ply of V-T-285.

3.3.6 Miscellaneous materials.

3.3.6.1 Heel fillers. The heel fillers shall be made of fiberboard or other suitable material that does not affect vulcanization or proper molding of the heel. The fillers shall be heel-shaped with straight edges. The heel fillers shall conform to the dimensions in table IV when tested as specified in 4.4.1.

TABLE IV. Dimensions of heel fillers in inches

Heel filler size designation	A	B	C	D
Heel curve radius	7/16	9/16	11/16	13/16
Width	7/8	1-1/8	1-3/8	1-5/8
Length	1-3/4	2	2-1/4	2-3/8
Thickness	9/16	9/16	9/16	9/16
Tolerance	± 1/32	± 1/32	± 1/32	± 1/32

3.3.6.2 Adhesives. The adhesives used for bonding the various leather and fabric components of the boots shall be of the following types:

a. Natural rubber latex
b. Synthetic rubber latex (including chloroprene)
c. Natural rubber solvent cement
d. Synthetic rubber solvent cement
e. Synthetic resin cement

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3.3.6.3 Instruction tags. Instruction tags, to be tied to the finished boots, shall be printed using the data supplied in figure 1. In addition, the instruction tags shall be stamped in red with the words "THIS BOOT IS SPIKE PROTECTIVE" across the top front. The tags shall be printed on tag stock that is 5-1/4 by 6-1/2 inches and folded in the middle to form four printed pages in book form that are 5-1/4 by 3-1/4 inches. A punch hole shall be made in the top near the folded edge for insertion of the lace.

3.3.6.4 Laces. The laces shall conform to type II, class 3 of V-L-61. The color of the laces shall be black. The length of each lace shall be a minimum of 60 inches.

3.3.6.5 Insoles, ventilating. Each pair of boots shall be supplied with one pair of ventilating insoles conforming to type II of MIL-I-1821. The ventilating insole size relation to applicable size boot shall be as indicated in MIL-I-1821.

3.3.6.6 Finishing materials for the leather uppers.

3.3.6.6.1 Repairers. Repairers shall be liquid spray, or crayon or paste type applied by hand. The color shall match the upper leather and shall have sufficient coverage to correct minor surface imperfections of the leather.

3.3.6.6.2 Fillers. Fillers, where used, shall be capable of providing a suitable foundation for the application of the top finish and shall be applied by sponge or spray method.

3.3.6.6.3 Top finish. Top finish shall be applied by either the sponge or spray method. It shall be a material compatible with and capable of wetting into and binding the previously applied coats. The finish shall match the dyed leather and shall dry to a low luster.

3.4 Design. The design shall be for a full lace closure, blucher pattern boot with integral steel sole plate and removable ventilating insole. The leather areas on the outside of the boot shall have the grain surface exposed. The boot shall have cotton and nylon cloth quarter with nylon webbing for backstay, top stay, and ankle reinforcements. The vamp shall be plain with a semihard box toe. The outersole design shall be bars and cleats.

* 3.5 Patterns and dies. The height of the finished boot, measured upward on outside from tread area at breast of heel to top of boot, shall be 9 7/8 inches ± 1/4 inch on size 9R and shall graduate up or down between sizes and widths as indicated by patterns. A standard set of paper patterns and markers will be furnished by the Government as a basis from which the contractor's dies and patterns shall be made. The patterns, markers, and contractor's cutting dies and patterns shall consist of the component parts specified in table V. The Government patterns and markers shall not be altered in any way

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and shall be followed, except for the steel insole plate (see 3.7.8.1.1) and for lasting allowance, which shall be determined by the contractor to provide a minimum of 1/2-inch allowance. Patterns and markers may be slightly modified to aid in fitting during manufacturing provided pattern part configuration or marker lap seams are not adversely altered dimensionally. An outside counterpocket may be used with an inverted "V" removed in the center of the counterpocket. The height of the cut for this inverted "V" may be as much as 1-1/4 inches from the bottom of the lasting allowance on a size 3 and grading upward to a maximum height of 1-1/2 inches on a size 14. The point of this inverted "V" shall be in the center back of the counterpocket. All outer lines of this counterpocket must conform to Government furnished patterns except for this inverted "V." This "V" shall be removed in a curved line to conform, after closing, to the back part of the MIL-5 last.

TABLE V. Contractor's dies and patterns

Component parts	Sizes
Vamp, full toe with nicks for box toe replacement	Whole and half sizes, all widths
Quarter	Whole and half sizes, all widths
Counterpocket inside	Whole and half sizes, all widths
Counterpocket outside	Whole and half sizes, all widths
Quarter front facing outside	Whole and half sizes, all widths
Eyelet stay, inside	Whole and half sizes, all widths
Gusset, bellows	Whole and half sizes, all widths
Box toe	8 sizes, all widths
Vamp lining	Whole and half sizes, all widths
Insole	Whole and half sizes, all widths
Steel insole plate	Whole sizes, all widths
Markers	
Vamp	Whole and half sizes, all widths
Quarter, with markers for attaching ankle reinforcement tape	Whole and half sizes, all widths
Gusset, bellows	Whole and half sizes, all widths

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3.6 Lasts and molds.

3.6.1 Lasts. The boots shall be lasted on the MIL-5 lasts in MIL-L-41815. All necessary sizes will be loaned to the contractor by the Government.

3.6.2 Molds. The molds shall be made so as to produce an outersole conforming to Drawing 2-1-962 when tested as specified in 4.4.2.1.

3.7 Construction.

3.7.1 Cutting uppers.

3.7.1.1 Cutting leather uppers. The leather components of the uppers shall be cut from grain out leather. The leather shall conform to the thickness specified in 3.3.1.1. Except for the front edge stays, eyelet stays, and inside counterpocket, none of the upper parts shall be split or shaved. Splitting of the outside counterpocket is optional. No stretchy vamps or parts cut off stretch shall be used. Vamps shall be cut from the bend area of the side leather.

3.7.1.2 Cutting cotton and nylon duck uppers. The cotton and nylon duck quarters shall be cut in the warp direction as indicated in figure 2. The cotton and nylon duck gusset may be cut in the warp, filling, or bias direction.

3.7.2 Skiving. The skiving of upper leather parts shall conform to requirements specified in table VI. The eyelet stays, front edge stays, and inside counterpocket shall be split or shaved to the thickness specified in table II. Splitting of the outside counterpocket is optional.

TABLE VI. Skiving requirements

Part	Side	Location	Scarf width in inches	Scarf edge thickness in millimeters (± 0.3 mm)
Vamp	Flesh	Throat and wings	1/4 ± 1/16	1.0
Vamp	Flesh	Toe 1/	3/4 ± 1/8	1.5

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TABLE VI. Skiving requirements - Continued

Part	Side	Location	Scarf width in inches	Scarf edge thickness in millimeters (± 0.3 mm)
Outside counter-pocket	Flesh	Stitch edge and top	3/16 ± 1/32	1.3
Outside counter-pocket	Flesh	Lasting edge 1/	5/8 ± 1/8	1.5

1/ Skiving of the lasting edge of the toe vamp and outside counterpocket is optional.

3.7.3 Crimping. The vamps may be crimped, if necessary, to ensure tight lasting.

3.7.4 Marking, permanent identification. The outersole shall be marked as shown on Drawing 2-1-962. The gusset of each boot shall be marked on the outside, approximately 1 inch from the top center of the gusset, with the correct size and width, the contractor's identification symbol in block form, the month and year (expressed numerically) of the date of contract, and the words, "SPIKE PROTECTIVE." Marking of case number is optional. The marking shall conform to type IV, class 9 of DDD-L-20. Fastness of the class marking shall be as specified for class 5 marking. Figures shall be Arabic and letters shall be Gothic. The figures and letters shall be a minimum of 9/32 inch and a maximum of 3/8 inch in height. The stamping shall be expressed as shown in the following example; alternate formats may be used provided all the data elements are included and requirements are met:

```
10 W   AB     6----81
     SPIKE      1084 (case number is optional)
  PROTECTIVE
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3.7.5 Upper fitting. Line marking patterns shall be used for all upper fitting. No die stab marking shall be allowed. The upper stitching shall be done with not less than 8 stitches nor more than 10 stitches to the inch. A plus tolerance of 3 stitches per inch will be allowed when stitching over heavy places or turning sharp corners, providing the leather is not damaged.

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Needles shall have a round point, except for vamping and barring, and shall be the proper size for the weight of leather and size of thread used. The back seam shall be joined using stitch type 401 of FED-STD-751. All other upper fitting shall be done on a machine using stitch type 301.

* 3.7.5.1 Fitting, ankle reinforcement tape, quarters, and counterpockets. When the 2-inch ankle reinforcement tape is used, the tape shall be positioned as indicated by the markers and shall be stitched to the exterior of the quarters with three single rows of stitches. Two of the single rows of stitches shall be positioned, one at each edge of the webbing, 1/16 to 3/32 inch from the edge. The third single row shall be positioned in the center of the webbing ± 1/16 inch. When the 1-inch ankle reinforcement tape is used, the two 1-inch pieces shall be positioned as indicated by markers and shall be stitched to the exterior of the quarters with four single rows of stitches. Four of the single rows of stitches shall be positioned, one at each edge of the 1-inch webbing, 1/16 to 3/32 inch from the edge. The ankle reinforcement tape shall then be trimmed even with the edges of the quarter. The nylon and cotton duck quarters shall be closed at the back with a closed seam 3/16 to 1/4 inch in width from top to bottom. This seam shall be rubbed down and reinforced with a stitched tape so that the tape will be covered by the backstay. The inside counterpocket shall be stitched, grain side out, to the marked position on the quarters with two rows on the edge. The top edge of the quarter shall be cemented and evenly folded outwards; the fold shall be 1/4 ± 3/32 inch in width. The outside counterpocket shall be stitched to the quarter marker position with two double rows of stitching. The individual rows of each double row shall be spaced 1/16 to 1/8 inch apart. The first double row of stitching shall be stitched through the cloth quarter 1/16 to 1/8 inch from the top edge of the outside counterpocket. The second double row of stitching shall be stitched through the outside counterpocket, the fabric quarter, and the inside counterpocket 3/8 ± 1/8 inch from the top edge of the outside counterpocket. The backstay, as specified in 3.3.2.6, shall be positioned 1/4 ± 3/32 inch from the top edge of the quarter and flush with the edge to plus 1/4 inch beyond the bottom edge of the quarter. The backstay webbing shall be stitched with a double row of stitching along both edges. The outer row of stitching shall be 1/16 to 3/32 inch from the edge. The individual rows of each double row shall be spaced 1/16 to 1/8 inch apart. The quarter top facing, as specified in 3.3.2.6 and cut in one piece, shall be placed even with the top edge of the quarter and stitched on the lower and upper edge. The quarter top facing shall overlay the backstay and extend under the outside quarter stay. The outside leather quarter stay shall be placed on the outside of the quarter, grain side out, extending a minimum of 5/16 inch beyond the quarter edge and under the counterpocket, then stitched with one row on the inside edge. If inverted "V" seam counterpocket is used, the "V" seam shall be closed, sewing upward from the lasting edge. The closing shall be done with one row of stitching using stitch type 401 and stitched with 8 to 10 stitches per inch. The seam shall extend approximately 1/16 inch on the flesh side and shall be rubbed down. The butt side of this seam shall be on the flesh side of the leather.

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* 3.7.5.2 Gusset fitting, eyelet stay attaching. The top of the nylon and cotton duck gusset shall be bound with the nylon tape specified in 3.3.2.5 and stitched with two rows of stitching. The first row shall be 3/32 ± 1/32 inch from the edge of the tape, and the second row shall be 3/32 ± 1/32 inch from the edge of the gusset; both of these rows of stitches shall be through the gusset. The bottom edge of the gusset shall be cemented and folded toward the inside, the folded part to measure 1/4 ± 3/32 inch in width. The inside leather eyelet stays shall be stitched, grain side out, at the marker position on the gusset with one row of stitching. The gusset shall be stitched to the vamp with one row of stitching not more than 1/8 inch from the edge of the gusset, and a second row placed not more than 5/32 inch from the edge of the vamp, both of these rows to include stitching through the vamp lining. The gusset shall be stitched to the quarters with two rows of stitching, one row on the outside edge that shall extend not less than 3/8 inch on the quarter top facing, and the second row on the inside eyelet stay that shall pass through the outer edge of the quarter panel fabric and gusset fabric without superimposing onto the eyelet stay edge row.

* 3.7.6 Eyeleting, vamping, and barring. On each quarter, there shall be eight eyelets for sizes 3 through 6-1/2, nine eyelets for sizes 7 through 11-1/2, and ten eyelets for sizes 12 through 14. The edge of each eyelet shall be 3/16 to 5/16 inch from the edge of the quarter. The eyelets shall be spaced evenly from blucher nose to top of quarter. The eyelets shall be smoothly and securely clinched. Vamping shall be done with two rows of stitching (single or double needle) close on the edge of the front edge stay, plus two rows spaced approximately 3/16 inch between inside rows. There shall be two single or double bar rows 5/8 ± 1/16 inch in length at each blucher ear. The bar row shall be placed outside and above vamping by 1/16 to 1/8 inch and parallel to the vamping row at blucher point. This barring may be done by the automatic method with tying or locking ends. Two drainage eyelets, as specified in 3.3.3.4.1, shall be inserted through the vamp lining on the inside shank of each boot and shall be located just above the insole on the finished boot. Two washers, as specified in 3.3.3.4.2, shall be used to clinch each drainage eyelet. The drainage eyelet shall be inserted with the flat side (perforated) on the grain side of the leather and the cup side innermost. The hole in the vamp through which the drainage eyelet is inserted shall be no more than 7/32 inch in diameter. As an alternative, two drainage washers, as specified in 3.3.3.4.2.1, shall be inserted through the vamp lining on the inside of each boot and shall be located just above the insole on the finished boot. One drainage eyelet, as specified in 3.3.3.4.1.1, shall be used to clinch each drainage washer. The drainage washer shall be inserted in the innermost side of the vamp and the drainage eyelet (perforated) on the grain side of the leather. The hole in the vamp through which the drainage washer is inserted shall be no more than 9/32 inch in diameter.

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3.7.7 Lacing for lasting. The machine lacing for lasting shall provide for an opening, after lasting, of 3/4 ± 1/8 inch. The quarter shall be laced in the three lower pairs of eyelets. As an alternative, the quarters may be laced in the first and third or the two lower pairs of eyelets.

3.7.8 Insoles. The insoles shall be sorted and cased for even weight. Insoles shall be fleshed and buffed on the flesh side. Die-cut insoles will be accepted, provided they conform to paper insole patterns loaned by the Government. A tolerance of +0 and -1/16 inch from Government loaned insole patterns will be allowed around the periphery of the insole.

3.7.8.1 Insole construction assembly. The insoles shall be assembled in three layers. The top layer, with grain surface next to the foot, shall be the 4-iron leather insole. The middle layer shall be the steel insole plate. The bottom layer shall be the 3-iron leather insole.

* 3.7.8.1.1 Steel insole plate and resistance to penetration. The steel insole plate, as specified in 3.3.3.5, shall be cut such that the distance between the outer edge of the plate and the outer edge of the leather insole shall be no more than 3/8 inch when assembled in accordance with 3.7.8.1.2. The Government patterns may be used for guidance (see 3.5). Both surfaces of the steel plate shall be nominally clean. The steel insole plate shall have a minimum average resistance to penetration of 215 pounds when tested as specified in 4.4.1.1. No single determination shall fall below 200 pounds.

3.7.8.1.2 Attaching leather insoles to steel insole plate. The leather insoles, as specified in 3.3.1.2, shall be cemented on the flesh surface of the 4-iron insole and on the top buffed or split surface of the 3-iron insole with any adhesive specified in 3.3.6.2. The leather insoles shall be attached to the steel insole plate, maintaining a margin of approximately 1/4 inch from the edge of the leather insoles to allow for stitching of the two attached layers of the leather insole, and making certain that the 4-iron leather insole grain surface will be against the foot (see 4.4.2.3.1). The assembly shall then be stitched using a polyester thread, with stitch type 101 or 301 of FED-STD-751. The stitching shall be 4 to 6 stitches per inch, shall be not less than 1/16 inch from the edge of the insole, and shall not penetrate the steel (see 4.4.2.3.2). When stitch type 101 is used, the chain shall be placed on the bottom surface of the assembly and not against the foot.

3.7.8.1.3 Insole tack holes and molding. If the insole tack holes are needed (see 3.7.9), prior to lasting, no more than four holes shall be drilled through the assembled insole. The holes shall be no more than 3/64 inch in diameter. The combined leather insole with steel plate may be molded to conform to the contour of the last bottom.

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3.7.9 Lasting. Insoles of the correct size and width shall be tacked to the last with four tacks. One of the four tacks shall be in the center of the heel seat, one at the shank, one in the ball, and one at the toe (see 3.7.8.1.3). As an alternative, the insole may be attached with four tacks on the outside 1/4-inch edge of the insole, so that two of the tacks are at the forepart and two are at the shank. Staples in lieu of tacks may be used for attaching the insole to the last. Care shall be taken that the edge of the insole is flush with or not more than 1/16 inch from the bottom edge of the last at all points and does not extend beyond the last at any point. The counters shall be assembled in accordance with size schedule so that they fill the counterpockets. The counter shall be seated to the insole and anchored with one or two tacks near the back end of the counter flange. The tack shall be driven through the outside counterpocket and insole and clinched against the steel insole plate. Steel tacks may be used for lasting operations. The counter shall be well cemented on both sides. When leather counters are used, the flesh side of the leather counter shall be toward the inside of the boot. The counter shall be turned over the last at the bottom 1/2 ± 1/16 inch. Uppers of the correct size and width shall be assembled to the last in a manner to provide for a wiped-in heel seat of not less than 7/16 inch nor more than 3/4 inch. The correct size of pyroxylin or polystyrene type box toe specified in 3.3.2.3 shall be conditioned and inserted between the vamp and vamp lining. A light coating of any adhesive specified in 3.3.6.2 shall be applied between the vamp and vamp lining rearward of the box toe. The uppers shall be drawn over the lasts with the proper tension on the pulling-over machine to ensure that the quarters at the blucher points and the vamps are down to the last and that the blucher noses are even. The heel seat shall be wiped-in and tacked, and shall be flat and free from wrinkles. The toe shall be firmly and smoothly wiped-in and securely attached by staples. The sides shall be cemented before or in conjunction with side lasting and stapled with a side stapling machine. The staples shall be adjusted and spaced to securely hold the upper in place through the vulcanization operation.

3.7.10 Time allowance on lasts. The boots shall remain on the lasts a sufficient length of time to thoroughly dry the counters and uppers.

3.7.11 Tack and staple pulling. All insole tacks and staples shall be removed with care taken that no broken tack or staple points remain.

3.7.12 Preparation of bottom. The bottom shall be prepared in accordance with the procedures detailed in 3.7.12.1 and 3.7.12.2.

3.7.12.1 Roughing. The edges of the lasted uppers shall be roughed with a cylindrical steel brush, 6.25 ± 0.25 inches in diameter and 1 or 2 inches in width, that is revolving at a speed of 2500 ± 400 revolutions per minute. The bristles shall have a diameter of 0.015 ± 0.003 inch. As an alternative, the bottom roughing may be performed with an automatic roughing machine or other suitable automatic roughing equipment. The first stage shall rough the outer

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edge (featherline) to the extent that the roughed surface will mold to the nip line of the outersole mold but will not be evident above the nip line. The second stage shall rough the remainder of the overlay in such a way as to remove the grain and leave the fibers rough and loose. Wheel roughing prior to wire brush roughing may be performed. Care shall be taken not to rough through the overlay. All loose particles of dust shall be completely removed from the roughed bottom with the use of a stiff bristle brush or compressed air.

3.7.12.2 Skiving and roughing. As an alternative to the procedure detailed in 3.7.12.1, the lasted grain side overlay portion of the counterpocket shall be skived in a sufficient width to ensure that the skived surface will mold to the nip line of the outersole mold but will not be evident above the nip line. The scarf edge thickness shall be 1.5 ± 0.3 mm. The entire skived edges, except the area where the stitches overlay the insole, shall be roughed in accordance with 3.7.12.1. The overlay portion of the vamp area shall be roughed in accordance with 3.7.12.1.

NOTE: When this alternative method is used, the skiving of the flesh side of the counterpocket shall not be permitted.

3.7.13 Preparation of upper for molding.

3.7.13.1 For biscuit method. A liberal amount of adhesive shall be applied with a hand brush to the roughed area and to the exposed part of the insole and the shank. After allowing the adhesive to dry for a minimum of 15 minutes, a second coat of the same adhesive shall be applied to the same areas and allowed to dry for a minimum of 40 minutes.

3.7.13.2 For injection method. The adhesive shall be applied in such a manner as to ensure compliance with the bond strength requirements specified in 3.3.4.

3.7.14 Attaching steel shank. The proper size of steel shank shall be attached to the upper while the boot is either on the MIL-5 last or on the metal last. The forward end of the steel shank shall be approximately 1-1/8 inches rearward of the ball line.

3.7.15 Relasting. The uppers shall be relasted using the proper size and width of the metal lasts on the direct molded sole machine.

3.7.16 Use of mold release. Coating with a mold release compound will be permitted on those surfaces of the metal mold that will come into contact with the soling compound.

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3.7.17 Molding outersole and heel to upper. The outersole and heel shall be direct molded to the upper by either the biscuit method or the injection method (see 6.6), using molding equipment that is approved for this purpose by the contracting officer. The metal lasts used in the molding operation shall conform to the measurements contained in MIL-L-41815. The rubber soling compound shall be as specified in 3.3.4. The heel fillers shall be as specified in 3.3.6.1 in four sizes (A, B, C, and D) and shall be applied to the boots in accordance with the schedule in table VII.

TABLE VII. Heel filler size schedule

Boot size	Boot width XN	N	R	W	XW
3, 3 1/2	-	-	-	-	A
4, 4 1/2	-	A	A	A	A
5, 5 1/2, 6, 6 1/2,	A	A	A	A	A
7, 7 1/2, 8, 8 1/2, 9, 9 1/2	B	B	B	B	B
10, 10 1/2, 11, 11 1/2, 12, 12 1/2	C	C	C	C	C
13, 13 1/2, 14	D	D	D	D	D

3.7.17.1 Biscuit molding method. The biscuits for the outersole and heel sections shall be cut from the same batch of rubber. The outersole biscuit shall extend from the toe to the rear of the heel in one unbroken piece. The heel biscuit shall be made from one or more strips of suitable thickness. A cavity for the heel filler shall be formed or cut in the heel biscuit to allow for a minimum margin of 5/8 inch between the cavity and the outer heel perimeter contacting the lasted upper, not including the heel breast line. The heel biscuit and the inner surface of the outersole biscuit may, when necessary, be lightly brushed with methyl ethyl ketone.

3.7.17.1.1 Alternate biscuit molding method. The outersole biscuit, including the heel, shall be cut from a calendered or extruded single biscuit of suitable thickness to fill the outersole and heel cavity of the mold. The heel filler shall be securely keyed to the steel shank, allowing a minimum margin of 5/8 inch between the periphery of the heel filler and the outer perimeter of the lasted upper.

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3.7.17.1.2 Biscuit molding conditions. The biscuit molding equipment shall be operated so as to exert a minimum pressure of 250 pounds per square inch (psi) on the compound. The sole and side molds shall be heated to a constant temperature of 300° to 325°F and shall hold the soling compound against the upper for vulcanization for a dwell time of 15 minutes minimum (see 4.4.2).

3.7.17.2 Injection molding method. The heel filler shall be securely keyed to the steel shank, allowing a minimum margin of 5/8 inch between the periphery of the heel filler and the outer perimeter of the lasted upper. The upper shall be mechanically held against the outersole and the heel mold, and the soling compound shall be injected into the mold under appropriate heat and pressure. The upper shall be held against the mold until the outersole is properly formed and cured.

3.7.18 Heel and edge scouring. After removal from the direct molded sole machine, the flash edges shall be removed when necessary by scouring. The use of a knife for trimming spewed flash is prohibited at the upper leather edge.

3.7.19 Tacks or staples. Tacks, nails, and staples that have been left protruding through or around the insole and cannot be pulled out shall be cut close to the surface and shall leave no protruding stumps.

3.7.20 Upper finishing.

3.7.20.1 Preparation. The boots shall be cleaned by removing accumulated dirt, such as wax or cement, and the surface of the leather shall be conditioned to receive further application of the finish. All thread ends shall be trimmed.

3.7.20.2 Treeing. All wrinkles shall be removed. No material shall be used that may injure the leather, fabric, or thread.

3.7.20.3 Final finish. The boots shall be repaired and filled. The materials and methods shall be as specified in 3.3.6.6.1 through 3.3.6.6.3.

3.7.21 Lacing. The boots shall be mated. A lace shall be inserted through the top eyelet of the outside quarter of each boot, and the two laces shall be tied firmly together.

3.7.22 Instruction tag. One instruction tag containing the information as shown in figure 1 shall be attached to each pair of boots.

3.7.23 Ventilating insole. One ventilating insole, as described in 3.3.6.5, shall be inserted in each boot.

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3.7.24 Repairs of stitching.

3.7.24.1 Types 301 and 401 upper fitting stitches. Repairs of types 301 and 401 stitches shall be as follows:

a. When thread breaks or bobbin run-outs occur during stitching, the stitching shall be repaired by restarting the stitching one stitch in back of the end of the secure stitching, except that on the quarter top facing, backstay, ankle reinforcement tape, and binding tape the stitching shall be restarted not less than 1/2 inch in back of the end of the secure stitching.

b. Thread breaks or two or more consecutive skipped or runoff stitches noted during inspection of the item (in-process or end item) shall be repaired by overstitching. The stitches shall start one stitch in back of the defective area and continue one stitch beyond the defective area onto the existing stitching, except that on the quarter top facing, backstay, ankle reinforcement tape, and binding tape the stitching shall be not less than 1/2 inch in each direction beyond the defective stitching area.

c. Vamp stitching damaged or missing at the junction of the outersole shall be repaired with a minimum of two stitches. Applicable to each row of defective stitching, the first stitch shall be a maximum of 1/4 inch from the junction of the outersole.

NOTE: In lieu of repairing each row of stitching individually, the four rows of stitching in the vamp may be repaired in one operation by stitching across the rows of stitching. Location of stitching shall be as specified above and shall start and end within the rows of stitching. Each group of four rows of stitching shall be repaired separately.

3.7.25 Replacement of defective components. During the spreading, cutting, and manufacturing process, components having material defects or damages that are classified as defects in 4.4.2.2.1, 4.4.2.2.2, and 4.4.3 shall be removed from production and replaced with nondefective and properly matched components.

3.8 Workmanship. The finished boots shall be free of defects, such as missing components, incorrect stitching patterns, unspecified materials, unmatched pairs, crooked box toe, unspecified color, wrong type or placement of hardware components, protruding points in insole, lack of ventilating insole, or missing instruction label. The boots shall also be free from stains or foreign matter affecting appearance.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirement in the document shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for assuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point or at all points in the manufacturing process necessary to assure compliance with all dimensional requirements.

4.1.3 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in 4.4.3 and tested for the characteristics specified in 4.4.4. The presence of any defect or the failure of any test shall be cause for rejection of the first article.

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4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

* 4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document. In addition, testing shall be performed on components and materials listed in table VIII. When the data items in the "Number of determinations per sample unit" and the "Results reported as" columns are not specified in table VIII, they shall be reported as required by the referenced test methods. A rubber compound lot shall consist of the amount of rubber compound that is produced or compounded in one day. A representative composite test sample of the lot of rubber used for the DMS biscuit method shall be vulcanized by using a 10-minute cure at 310°F. A representative composite test sample of the lot of rubber used in the injection molding method shall be molded and cured at the same temperature and time as that used in the production of the boots. The composite test sample shall be as follows:

a. Two pieces, 6 inches by 6 inches by 0.070 ± 0.010 inch

b. Two pieces, 6 inches by 6 inches by 0.250 ± 0.010 inch

All test reports shall contain the individual values utilized in expressing the final result. The lot shall be unacceptable if one or more sample units or the composite sample fails to meet any specified requirement except for abrasive index. For abrasive index, all composite samples from the rubber compound lots making up the boot lot shall be tested. If the average of all composite samples fails the abrasive index, the boot lot shall be rejected. The sample size shall be as follows:

Lot size	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

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TABLE VIII. Component tests

Component and lot size expressed in terms of	Characteristic	Requirement paragraph	Test method	Requirement applicable to: Sample unit	Requirement applicable to: Composite	No. determinations per sample unit	Results reported as: Pass or fail	Results reported as: Numerically to nearest	Sample unit
Box toes (1 box toe)	Material identification	3.3.2.3	1/	-	-	-	-	-	-
	Impregnation compound identification	3.3.2.3	1/	-	-	-	-	-	-
	Thickness	3.3.2.3	1/	-	-	-	-	-	-
Nylon binding tape (gross yd)	Colorfastness to weathering	3.3.2.5.1	5671 2/3/	X	-	-	-	-	2 yards
Nylon webbing reinf. tape (gross yd)	Colorfastness to weathering	3.3.2.6	5671 2/3/	X	-	-	-	-	1 yard

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TABLE VIII. Component tests - Continued

Component and lot size expressed in terms of	Characteristic	Requirement paragraph	Test method	Requirement applicable to: Sample unit	Requirement applicable to: Composite	No. determinations per sample unit	Results reported as: Pass or fail	Results reported as: Numerically to nearest	Sample unit
Shank, steel (1 pair)	Material identification	3.3.3.1	1/	-	-	-	-	-	-
	Thickness	3.3.3.1	1/	-	-	-	-	-	-
	Hardness (Rockwell C scale)	3.3.3.1	E 18 4/	X	-	1	-	Hardness number	-
	Height, steel rib	3.3.3.1	Visual, micrometer	X	-	1	X	-	2 shanks
	Length and width	3.3.3.1	1/	-	-	-	-	-	-
	Coating identification	3.3.3.1	1/	-	-	-	-	-	-
	Finish	3.3.3.1	1/	-	-	-	-	-	-
Tacks and staples	Material identification	3.3.3.2	1/	-	-	-	-	-	-

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TABLE VIII. Component tests - Continued

Component and lot size expressed in terms of	Characteristic	Requirement paragraph	Test method	Requirement applicable to: Sample unit	Requirement applicable to: Composite	No. determinations per sample unit	Results reported as: Pass or fail	Results reported as: Numerically to nearest	Sample unit
Eyelets 1 pound	Material identification	3.3.3.3	1/	-	-	-	-	-	-
	Outside diameter of flange and barrel	3.3.3.3	Gage	X	-	2	-	0.001 inch	-
	Diameter of hole	3.3.3.3	Gage	X	-	2	-	0.001 inch	2 eyelets
	Overall length	3.3.3.3	Gage	X	-	2	-	0.001 inch	-
	Thickness	3.3.3.3	1/	-	-	-	-	-	-
	Finish	3.3.3.3.1	1/	-	-	-	-	-	-
Drainage eyelets 1 pound	Diameter head	3.3.3.4.1	Gage	X	-	2	X	-	-
	Outside diameter of barrel	3.3.3.4.1	Gage	X	-	2	X	-	-

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TABLE VIII. Component tests - Continued

Component and lot size expressed in terms of	Characteristic	Requirement paragraph	Test method	Requirement applicable to: Sample unit	Requirement applicable to: Composite	No. determinations per sample unit	Results reported as: Pass or fail	Results reported as: Numerically to nearest	Sample unit
Drainage eyelets 1 pound (continued)	Length of barrel	3.3.3.4.1	Gage	X	-	2	X	-	2 drainage eyelets
	Outside diameter	3.3.3.4.1.1	Gage	X	-	2	X	-	-
	Length overall	3.3.3.4.1.1	Gage	X	-	2	X	-	-
Washers	Outside diameter	3.3.3.4.2	Gage	X	-	2	X	-	-
	Inside diameter	3.3.3.4.2	Gage	X	-	2	X	-	-
	Thickness	3.3.3.4.2	Gage	X	-	2	X	-	-
	Head diameter	3.3.3.4.2.1	Gage	X	-	2	X	-	2 washers
	Outside diameter of barrel	3.3.3.4.2.1	Gage	X	-	2	X	-	-
	Length of barrel	3.3.3.4.2.1	Gage	X	-	2	X	-	-
	Thickness	3.3.3.4.2.1	Gage	X	-	2	X	-	-

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TABLE VIII. Component tests - Continued

Component and lot size expressed in terms of	Characteristic	Requirement paragraph	Test method	Requirement applicable to: Sample unit	Requirement applicable to: Composite	No. determinations per sample unit	Results reported as: Pass or fail	Results reported as: Numerically to nearest	Sample unit
Outersole and heel compound (lot)	Volume swell	3.3.4	6001 and 6211 5/	-	X	3	-	One percent	-
	Abrasive index (before and after aging)	3.3.4	D 1630 4/ 6/ 7/	-	X	3	-	Whole number	- -
	Hardness (before and after aging)	3.3.4	D 2240 4/ 7/	-	X	5	-	Whole number	-
	Cut growth (after aging)	3.3.4	D 1052 4/ 6/	-	X	2	-	One percent	- -
	Ozone resistance	3.3.4	4.5.1 1/	-	X	1	X	-	-
Heel filler (1 filler)	Heel curve radius	3.3.6.1	Gage	X	-	2	X	-	-
	Width	3.3.6.1	Gage	X	-	2	X	-	2 fillers of each size in the lot
	Length	3.3.6.1	Gage	X	-	2	X	-	
	Thickness	3.3.6.1	Gage	X	-	2	X	-	

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1/ A certificate of compliance shall be submitted and will be acceptable for the stated requirement.

2/ Refers to FED-STD-191.

3/ The time of exposure shall be 40 standard hours.

4/ Refers to ASTM Standard.

5/ Refers to FED-STD-601.

6/ Aging shall be at a temperature of 100° ± 1°C for 70 hours in accordance with the ASTM D 573 procedure for accelerated aging.

7/ The abrasive paper to be obtained from the Carborundum Co. shall be G40E155F.

4.4.1.1 Testing of steel insole plate. The steel insole plate shall be tested as specified in 4.5.2 for conformance to the resistance to penetration requirements specified in 3.7.8.1.1. Fifteen insole plates shall be selected at random from the lot and tested. Any test failure shall be cause for rejection of the lot of insole plates.

4.4.2 In-process inspection. Inspection shall be made at any point or during any phase of manufacturing to determine whether operations or assemblies are accomplished as specified. When the DMS biscuit method is used, the contractor shall furnish a certificate of compliance for the outersole and heel molding as specified in 3.7.17.1.2 for pressure, heat, and dwell time. The Government reserves the right to exclude from consideration for acceptance any material or service for which in-process inspection has indicated nonconformance.

4.4.2.1 Mold approval. During the molding operation and prior to the formation of the initial inspection lot, one boot shall be randomly selected from the production of each new mold and determination shall be made whether the outersole conforms to the measurements specified on Drawing 2-1-962. If the outersole is found to be in conformance, the mold from which it was made shall be approved. If the measurements are found to be nonconforming, an additional five boot outersoles produced from that mold shall be measured for conformance to Drawing 2-1-962 and, if all five outersoles conform to the measurements, the mold shall be approved. If all molds used in the processing of the initial inspection lot have been approved, testing of the finished boots for conformance to outersole measurements shall not be applicable. When new molds are added or when molds are repaired, mold approval must be obtained as described previously.

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4.4.2.1.1 Identification marking of approved molds. Each mold after approval shall have an identifying number inscribed approximately in the center of the heel breast portion of the heel mold. The mold identification numbering shall be consecutive, beginning with number "1." The height of the numbers shall be not more than 1/8 inch.

4.4.2.2 In-process examination. The defects found during in-process examination shall be classified in accordance with 4.4.2.2.1 and 4.4.2.2.2. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major and 6.5 for total (major and minor combined) defects.

NOTE: Defects designated by an asterisk shall be scored as major when seriously affecting serviceability and minor when affecting serviceability but not seriously.

4.4.2.2.1 Examination of uppers after all fitting. The completely fabricated upper assembly prepared for lasting shall be examined for defects that cannot be seen in the end item, as listed in table IX. The lot size shall be expressed in units of one upper assembly. The sample unit shall be one upper assembly.

TABLE IX. Upper defects

Examine	Defect	Classification	
		Major	Minor
Construction and workmanship (general)	Any component missing or not specified type	X	
	Any component misplaced or not affixed as specified	*	
	Vamp throat or wings not skived	X	
	Counterpocket not skived as specified		X
	Vamp stitched more than 5/32 inch from edge at throat	X	
	Vamp stitched more than 3/32 inch from edge at throat but not over 5/32 inch		X
	Backseam not rubbed down		X

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TABLE IX. Upper defects - Continued

Examine	Defect	Classification	
		Major	Minor
Quality of leather	Thickness more than 1/2 ounce less than the minimum specified	X	
	Thickness less than specified minimum but not exceeding 1/2 ounce		X
	Thickness more than the maximum specified		X
	Off-stretch cut	X	
	Slaughter cut	*	

4.4.2.2.2 Examination of boot before last pulling. The partially fabricated boot assembled to the point just prior to pulling off the MIL-5 last shall be examined for defects that cannot be seen in the end item, as listed in table X. The lot size shall be expressed in the units of one partially fabricated boot. The sample unit shall be one partially fabricated boot.

TABLE X. Boot before last pulling defects

Examine	Defect	Classification	
		Major	Minor
Bottom of boot	Any component missing or not specified type	X	
	Upper damaged	*	
	Insole not fleshed and buffed on the flesh side as required	X	
	Poor heel seat, side, or toe lasting	*	

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TABLE X. Boot before last pulling defects - Continued

Examine	Defect	Classification	
		Major	Minor
Upper part of boot	Upper not tightly pulled down to last		*
	Lace opening less than 5/8 inch or more than 7/8 inch		X
	Quarter not laced as specified		X
	Less than 1/2 inch of upper lasted over insole except heel seat upper less than 7/16 inch or more than 3/4 inch	X	
	Upper in lasted area not sufficiently roughed or cleaned after roughing	X	
	Excessive roughing or scouring		*

4.4.2.3 Examination of the assembled insole.

4.4.2.3.1 Steel plate. Each assembled insole shall be examined to ensure that the steel plate has not been omitted. A mechanical or manual metal detector or other suitable automatic metal detecting equipment shall be used. As an alternative to metal detecting equipment, the assembled insole may be hand flexed to ensure that the plate has not been omitted. Any assembled insole that does not include the steel insole plate shall be rejected.

* 4.4.2.3.2 Insole assembly. The assembled insole shall be examined for defects listed in table XI. The lot size shall be expressed in units of one assembled insole. The sample unit shall be one assembled insole. The inspection level shall be II, and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

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TABLE XI. Insole defects

Examine	Defect
Insole stitching	More or less than the specified number of stitches Wrong stitch type Less than 1/16 inch from the edge of the insole Stitches penetrating steel insole Skipped or broken stitch
Steel plate	Distance between outer edge of steel plate and outer edge of leather insole more than 3/8 inch (examined at ball and toe areas)

* 4.4.3 End item visual examination. The finished boots shall be examined for the defects in table XII. The lot size shall be expressed in units of one boot. The sample unit shall be one completely fabricated boot, and the selection shall be by pairs. The inspector shall check to see that ventilating insoles are inserted in the boots. For the pairing examination and when determining a possible difference in outersole thickness, the pair shall be examined together. Each defect found during the examination for pairing shall be scored as a single defect. The inspection level shall be II, and the AQL, expressed in terms of defects per hundred units, shall be 2.5 for major and 6.5 for total (major and minor combined) defects. To examine the vamp for break, the boot shall be held in an upright position with both hands. The toe of the boot shall face away from the examiner. Position thumbs on top of vamp approximately halfway between box toe line and blucher noses and 1 inch to 1-1/2 inches apart. Press downward with thumbs so as to form grain surfaces into a concave surface and observe the break pattern. Any vamp exhibiting a break pattern at any location comparable to a break pattern represented by numbers 5 through 8 on the Satra scale (see 6.7) shall be scored as a defect. The defect shall be scored regardless of the direction in which the break pattern appears.

NOTE: Appearance defects shall be scored only when the condition is plainly visible at a distance of 3 feet or more. Defects designated by an asterisk shall be scored as major when seriously affecting serviceability or appearance and minor when affecting serviceability or appearance but not seriously.

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TABLE XII. End item visual defects

Examine	Defect	Classification	
		Major	Minor
Pairing	Not properly mated (i.e., not right and left of same size)	X	
	Variation in color, luster, or appearance		*
	Variation of more than 1/4 inch in height of pair	X	
	Box toe crooked, long, or short		*
	Difference in outersole thickness between left and right boots:		
	-more than 3/32 inch	X	
	-more than 1/16 inch but not more than 3/32 inch		X
Cleanness	Any nonremovable spot, stain, or foreign matter affecting appearance		X
Color and finish	Not specified color	X	
	Finish streaky, chipped, or flaky on upper		X
Design, type, and size	Not as specified	X	
Upper leather	Leather deeply snuffed (i.e., fiber structure damaged)	X	
	Any grub or tick damage, scratch, brand, or boney leather		*
	Fat wrinkles, flanky leather, or veins	X	
	Stretchy vamp	X	
	Leather embossed or printed	X	
Break pattern of vamp	Break pattern comparable to break pattern represented by numbers 5 through 8 on Satra break scale	X	

NOTE: Leather exhibiting a break pattern comparable to the numbers indicated above is considered to be loose, pipey, or flanky. The break pattern shall be determined in accordance with 4.4.3.

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TABLE XII. End item visual defects - Continued

Examine	Defect	Classification	
		Major	Minor
Nylon and cotton quarter and gusset	Any mend or weak spot	X	
	Two or more yarns missing	X	
	Any quarter cut on the bias	X	
	Any abrasion, spot, stain, or discoloration		*
Construction and workmanship	Any cut, tear, hole, repair, abrasion, or factory damage		*
	Any component or assembly omitted or misplaced, operation omitted or not properly performed (unless otherwise classified herein)		*
Webbing	Not firmly and tightly woven, edges frayed or scalloped	X	
Seams and stitching (upper)	Open seam not repaired	X	
	Repair of open seam not as specified		X
	Quarter panel fabric and gusset fabric not caught in second row of inside eyelet stay stitching		X
	NOTE: A seam shall be classified as open when two or more consecutive skipped or run-off stitches occur; or when one or more stitches joining a seam are broken, except that where vamp stitching crosses the counterpocket stitching area or the quarter top facing, backstay, ankle reinforcement and binding tape, two or more broken stitches shall constitute an open seam. On multiple stitched seams, a seam is considered open when either one side or both sides of seams are open. When any of the above conditions occur on the inside of the boot, it shall be scored as an asterisk (*) defect.		

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TABLE XII. End item visual defects - Continued

Examine	Defect	Classification	
		Major	Minor
Vamp stitching at outersole junction	One or more rows of defective stitching not repaired	X	
	Repaired but first stitch more than 1/4 inch from outersole junction	X	
	Repaired with one stitch in lieu of two or more stitches		X
	Tight tension resulting in puckering or cutting of leather or fabric	*	
	Loose tension resulting in a loosely secured seam	*	
	Wrong stitch or seam type	X	
	Any row of stitching with less than 8 stitches per inch but not less than 6 stitches per inch		X
	Any row of stitching with less than 6 stitches per inch	X	
	More than the specified maximum number of stitches	X	
NOTE:	A plus tolerance of 3 stitches per inch will be allowed when stitching over heavy places or turning sharp corners.		
	Gage of stitching not as specified or irregular	*	
	Stitching omitted where required	*	
	Thread ends not trimmed throughout boot		X
	Needle holes or needle chew	*	
	Sand hole (i.e., tongue not properly caught in at throat)		X

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TABLE XII. End item visual defects - Continued

Examine	Defect	Classification	
		Major	Minor
Outersole and heel	Thickness not as specified	X	
	Flash edges not removed by scouring, affecting appearance		X
	Edge scouring is irregular, affecting appearance		X
	Blister:		
	-1/2 inch or more in longest dimension	X	
	-penetrating 1/3 or more the thickness of the rubber	X	
	More than three blisters more than 1/8 inch but less than 1/2 inch in longest dimension	X	
	One, two, or three blisters more than 1/8 inch but less than 1/2 inch in longest dimension		X
	Any void or knit mark 3/4 inch or more in longest dimension	X	
	Any void or knit mark penetrating 1/3 or more the thickness of the rubber	X	
	More than one void or knit mark in the outside sole lugs or rear heel lug, 1/2 inch or more in longest dimension	X	
	More than three voids or knit marks in the outersole and heel, 1/2 inch or more in longest dimension	X	
	One, two, or three voids or knit marks 1/2 inch up to but not including 3/4 inch in longest dimension		X
	More than two voids or knit marks more than 1/4 inch but less than 1/2 inch in longest dimension		X
	Any other defect in tread or edge area	*	
	Incorrect or incomplete vulcanization of outersole or heel to uppers (i.e., rubber not securely attached)	*	

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TABLE XII. End item visual defects - Continued

Examine	Defect	Classification	
		Major	Minor
Counters	Rolled or curled counter	*	
	Soft counter		X
Heel finishing	Heel not finished square and smooth, affecting appearance		X
Eyelets, quarter	Number of eyelets not as specified but each row has the same number		X
	Not the same number of eyelets in each row	X	
	Eyelets not properly spaced within the row of misalignment between the rows to an extent interfering with proper lacing	*	
	Edge of any eyelet less than 1/16 inch from edge of quarter	X	
	Edge of any eyelet less than 3/16 inch but not less than 1/16 inch or more than 5/16 inch from the edge of the quarter		X
	Any eyelet not securely clinched		X
Drainage eyelets and washers	Any drainage eyelet or washer missing	X	
	Not securely clinched, or not inserted as specified	*	
Gusset	Binding missing	X	
Barring	Missing	X	
Insole	Short or long	*	
	Any protruding point of lasting staple, or tack in heel 1/	X	

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TABLE XII. End item visual defects - Continued

Examine	Defect	Classification	
		Major	Minor
Fabric: quarters vamp lining, or ankle reinforcement tape	Vamp lining or quarter and ankle reinforcement tape having excessive fullness; torn, loose, or wrinkled	*	
Ventilating insole	Missing, wrong size, or wrong side up		X
Markings and instruction tag	Missing, incomplete, incorrect, not applied in the specified manner, misplaced, illegible, or not specified size		X

1/ Any protruding point of lasting staple or tack forward of heel breast line found in the sample shall be cause for rejection of the lot represented.

4.4.4 End item testing. The finished boot shall be tested for the characteristics listed in table XIII. The sample unit for all tests shall be one boot. All test results shall contain the individual values utilized in expressing the final result. The lot shall be unacceptable if one or more sample units fail to meet any requirement specified. The sample size shall be five boots regardless of lot size.

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TABLE XIII. End item tests

Characteristics	Requirement paragraph	Test method	No. determinations per sample unit	Results reported as Pass or fail	Results reported as Numerically to nearest
Nonmarking (outersole and heel)	3.3.4	4.5.3	1	X	-
Bond strength (outersole)	3.3.4	4.5.4	1	-	lb
Measurements 1/	Drawing 2-1-962	Gage	1	X	-

1/ Mold approval (see 4.4.2.1).

4.4.5 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components, such as incomplete sealing or closure of flap, improper taping, loose strapping, or inadequate stapling Bulged or distorted container
Content	Number per container is more or less than required

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4.4.6 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement
Palletization	Pallet pattern not as specified Interlocking of loads not as specified Load not bonded as specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

4.5 Methods of inspection.

4.5.1 Ozone resistance test. There shall be no sign of cracking of the outersole compound when tested as follows: Test specimens 1 inch in width, 3-3/4 inches in length, and 0.070 ± 0.010 inch in thickness shall be mounted in accordance with the requirements of ASTM D 518, procedure B, except that the length of the clamping strips shall be such as to facilitate placement within the test chamber of the ozone cabinet. Samples shall be exposed for 7 days to an ozone concentration of 50 ± 10 parts per hundred million parts of air at a temperature of 40° ± 1°C as described in ASTM D 1149. At the end of the exposure time, the specimens shall be examined under a seven-power magnifier.

4.5.2 Resistance to penetration test.

4.5.2.1 Apparatus. Resistance to penetration shall be performed on a testing machine having a movable platform adjusted to travel at 1/4 inch per minute. Two blocks of hard wood, metal, or plastic shall be prepared as follows: The blocks shall be of such size and thickness as to ensure a suitably rigid test ensemble and allow for at least 1 inch of the pointed end of an 8D common nail to be exposed for the penetration. One block shall have a hole drilled to hold an 8D common nail firmly at an angle of 90 degrees. The second block shall have a maximum 1/2-inch diameter hole drilled through it so that the hole will allow free passage of the nail after it penetrates the insole plate during the test.

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4.5.2.2 Procedure. Place a new 8D nail in the hole of the block, a sample of outersole stock 10 to 11 irons thick superimposed above the nail, the area of the insole plate to be tested on the outersole, and the second block with hole so as to allow for free passage of the nail after it passes through the outersole stock and insole plate in that order. Start the machine and record the pressure required for the nail to completely penetrate the outersole and insole plate, to the nearest 5 pounds. Make two determinations on each insole plate and average the results. A new nail shall be used for each determination. Any average reading below 215 pounds or any single determination falling below 200 pounds shall be considered a test failure.

4.5.3 Test for nonmarking outersoles and heels. The edge of the outersole or heel shall be drawn across a sheet of white bond paper using moderate pressure. If only a light mark is made that is easily erased by rubbing with the fingers, the outersole or heel shall be classed as nonmarking.

4.5.4 Bond strength test of outersole to leather.

4.5.4.1 Specimen. The test specimen shall be a single finished boot of any size prepared as follows: The boot shall be legibly marked on each side at two points (i.e., at 2 inches and 4 inches from the tip of the toe). The molded outersole shall then be separated from the upper to the 2-inch mark. The separation may be aided by any suitable device or machine capable of withstanding the amount of force necessary to accomplish separation. The use of pincers conforming to type II, class 1 of GGG-N-350 or equivalent, with the sharp cutting edge ground down to a 1/8-inch face, and a last one or two sizes smaller than the boot, inserted into the boot and mounted on a last stand, have been found to facilitate the preparation of specimens for the bond strength test. After separation to the 2-inch mark is obtained, the boot last, when used, shall be removed and the toe cap of the boot shall be crushed to facilitate mounting of the specimen in the tester. In the event that separation at the 2-inch mark is not between the upper and rubber outersole, that portion of the rubber outersole remaining adhered to the upper shall be separated manually with the aid of a knife or other instrument so as to show separation between leather and rubber outersole. Care should be exercised so that the leather is not cut.

4.5.4.2 Procedure. The apparatus for measuring outersole-upper separation shall be as described in method 2031 of FED-STD-311. Grip the separated and crushed toe portion of the boot by one clamp in a manner such that the minimum effective jaw surface area applied shall be 1 inch by 1 inch. (With some testing machines, modification of the clamps may be required to provide sufficient opening for insertion of the specimen.) Center the jaw surface approximately 1/2 to 1 inch from the tip of the toe. The other jaw surface shall hold the overlay portion of the separated upper and shall be not less than 1 inch by 2 inches with the long dimension perpendicular to the pull of the machine. Grip the separated rubber outersole portion of the boot by the other clamp of the testing machine. Each jaw of the clamp shall be not less

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than 1 inch by 2 inches with the long dimension perpendicular to the pull of the machine. Set the machine in operation, and continue the separation at a speed of 10 ± 2 inches per minute until the outersole is separated from the upper to a point past the mark 4 inches from the tip. The maximum value attained during separation shall be taken as the bond strength.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial, as specified (see 6.2).

5.1.1 Level A. Each pair of properly mated boots shall be tied together by the ends of the laces that have been inserted through the top eyelet of the outside quarter of each boot as specified in 3.7.21.

5.1.2 Commercial. Boots shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial, as specified (see 6.2).

5.2.1 Level A packing. Twelve pairs of boots of one size and width only, preserved as specified in 5.1, shall be arranged as described in 5.2.1.1 and packed in a fiberboard shipping container conforming to style FOL-L, grade V2s of PPP-B-636. The inside of each container shall be fitted with a box liner conforming to type CF, class weather-resistant, variety DW, grade V15c of PPP-B-636, except that the liner joint may be secured in accordance with commercial practice. The approximate inside dimensions of the shipping container shall be as specified in table XIV. Each container shall have the contents completely covered on the top and bottom with a sheet of 30-pound minimum basis weight kraft paper conforming to A-A-203. Each container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in PPP-B-636, except that the inspection shall be in accordance with 4.4.5. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). When unit loads are strapped, strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

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TABLE XIV. Approximate inside dimensions of shipping container 1/

Range of boot sizes	Length (inches)	Width (inches)	Depth (inches)
5 XN through 8 1/2 XW	30-1/2	12-1/2	13
9 XN through 12 XW	30-1/2	13-1/4	13-3/4

1/ Approximate dimensions are furnished as a guide only. For boot sizes below 5 XN and above 12 XW, the contractor shall determine the size of container to use. The dimensions of the container shall be such that it will result in a snug fit of the contents to be packed therein. The use of blocking materials to accommodate small size boots in an oversized container will not be permitted.

5.2.1.1 Packing arrangement. Boots within the shipping container shall be arranged in four layers of three pairs per layer on their sides, with the first layer started by placing the left boot in the left-hand end of the container with the heel toward the front and the sole facing left, and the right boot laid over the quarter of the left boot in the same manner. Two more pairs shall be placed in a like manner to complete the first layer. Care shall be taken that the first layer is properly spaced in order to facilitate the packing of the other three layers. The second layer shall start at the left by placing the right boot with the heel toward the rear and the top facing left and the sole face to face with the heel of the second boot in the first layer, and so on to complete the layer. The third layer shall be placed exactly as the first, and the fourth layer shall be the same as the second. The last boot placed in the first and third layers and the first boot placed in the second layer shall have the tops turned upward. The first boot placed in the top layer shall have the top folded under (see figure 3).

5.2.2 Level B packing. Twelve pairs of boots of one size and width only, preserved as specified in 5.1, shall be arranged as described in 5.2.1.1 and packed in a fiberboard shipping container conforming to style FOL-L, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. The manufacturer's joint shall be metal stitched. The inside of each fiberboard container shall be fitted with a box liner conforming to type CF, class domestic, variety DW, grade 275 of PPP-B-636, except that the liner joint may be secured in accordance with commercial practice. The approximate inside dimensions of the shipping container shall be as specified in table XIV. Each

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container shall have the contents completely covered on the top and bottom with a sheet of 30-pound minimum basis weight kraft paper conforming to A-A-203. Each container shall be closed in accordance with method II as specified in PPP-B-636 except that the inspection shall be in accordance with 4.4.5.

5.2.2.1 Weather-resistant fiberboard containers. When specified (see 6.2), the shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in PPP-B-636 except that the inspection shall be in accordance with 4.4.5.

5.2.3 Commercial packing. The boots, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

* 5.3 Palletization. When specified (see 6.2), boots, packed as specified in 5.2.2 and 5.2.3, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet type shall be type I (4-way entry), type IV or type V in accordance with MIL-STD-147. Pallets shall be fabricated from wood groups I, II, III, or IV of MIL-STD-731. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L or film bonding means O or P except that for bonding means P a fiberboard sheet on top of the load is not required. Pallet pattern shall be number 2 in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course.

5.4 Marking. In addition to any special marking required by the contract or purchase order, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

6.1 Intended use. These boots are intended for use in tropical areas by military personnel of the Department of Defense.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. Size and width required (see 1.2).
c. When a first article is required (see 3.2, 4.3, and 6.3).
d. Selection of applicable levels of preservation and packing (see 5.1 and 5.2).
e. Type and class of unit load required (see 5.2.1).
f. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
g. When palletization is required (see 5.3).

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* 6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for selection, inspection, and approval of the first article.

6.4 Standard sample. For access to samples, address the contracting activity issuing the invitation for bids.

6.5 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this document (see 3.3).

6.6 Direct molding methods for attaching soles (see 3.7.17).

6.6.1 Biscuit method. Slabs (biscuits) of compounded uncured rubber are placed in heated molds. The biscuits are applied against prepared uppers under heat and pressure necessary to form the outersoles and heels and to vulcanize them to the uppers.

6.6.2 Injection method. Prepared uppers are mechanically held against outersole and heel molds. The soling compound is then injected into the molds under heat and pressure and held against the uppers until the outersole and heel are formed and cured.

* 6.7 Satra scale. The Satra scale may be obtained from the British shoe and Allied Trade Research Association, Satra House, Kettering, Northants, England or may be obtained from Satra House's North American agent: Bata Engineering Batawa, Ontario, Canada.

6.8 International standardization agreements. Certain provisions of this document are the subject of international standardization agreement as cited in NATO, STANAG No. 2333. When amendment, revision, or cancellation of this document is proposed that will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization offices, if required.

* 6.9 Subject term (key word) listing.

Boots, hot weather
Climate, tropical
Clothing, tropical

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6.10 Changes from previous issue. The margins of this document are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content, as written, irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - NU
Air Force - 99

Review activities:

Army - MD
Navy - MC
Air Force - 82
DLA - CT

User activity:

Air Force - 45

Preparing activity:

Army - GL

Project No. 8430-0367

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INSTRUCTION TAG

BOOTS, HOT WEATHER

INSTRUCTIONS FOR USE

1. Wear with: 1 Pair socks, wool, cushion sole.
1 Pair insoles, ventilating.

2. Sizes. The boot is supplied in whole and half sizes 3 through 14 in all widths except sizes 3XN and 3 1/2XN, 3N and 3 1/2N, 3R and 3 1/2 , 3W and 3 1/2W, and 4XN and 4 1/2XN. In some instances it may be necessary to make a size adjustment in order to obtain an adequate fit due to normal swelling of the feet and the use of ventilating insoles.

3. Pull on boot, seating heel firmly into place, then double lace at third or fourth eyelet. Boot should fit snugly, but not tightly.

4. Trouser should be worn outside the boot.

5. Carry extra socks with you. Change socks often, at least once daily. Remove boots, and dry feet as often as possible. Neglect of the feet can cause Jungle Rot and make you a casualty.

6. Remove ventilating insoles, wash insoles in warm (not boiling) soapy water and shake out to dry.

7. Brush mud and excessive dust off boots. To prevent leather from getting hard and stiff, wax often with shoe polish containing silicone.

8. Clean screened vents so that water will drain out.

9. Wipe inside of boot occasionally with a moist soapy rag, and let dry.

10. As frequently as possible, dry boots and insoles in the sun. Do not place near intense heat as this will destroy nylon, leather, and plastic.

11. Before putting on boots in snake, insect, scorpion, or spider infested areas, carefully inspect the inside of the boots to avoid painful and disabling bites and stings.

FIGURE 1. Hot weather boots instruction tag.

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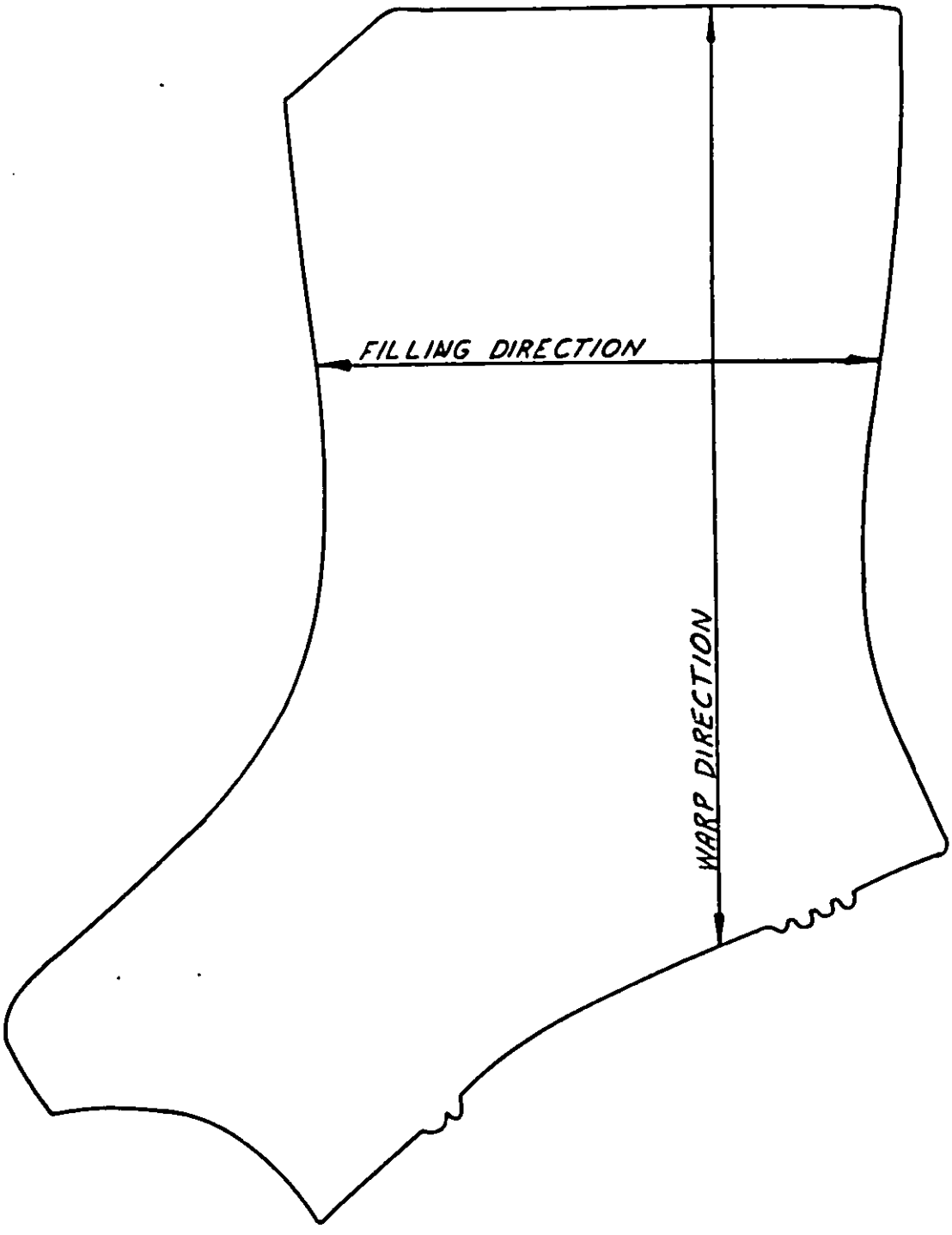


FIGURE 2. Boots, hot weather.

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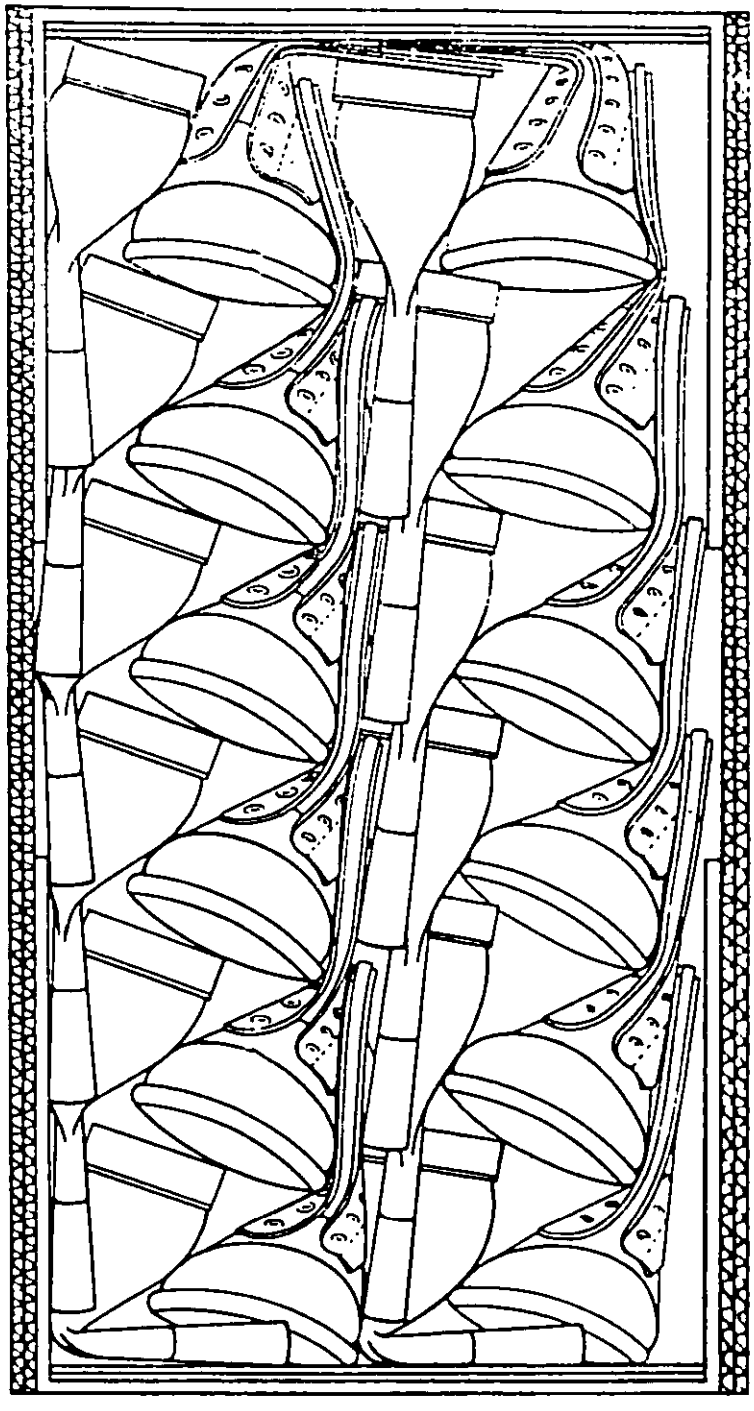


FIGURE 3. Boot arrangement within shipping container (side removed).

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-43154K	Boots, Hot Weather

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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